

AI Healthcare Assistant

Hugging Face and Prompt Engineering

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1. Introduction

Artificial Intelligence (AI) has become an important technology in the healthcare industry by assisting medical professionals in analyzing patient information, summarizing clinical reports, and answering healthcare-related queries. Large Language Models (LLMs) further enhance these capabilities by generating human-like responses based on natural language input.

This project develops an AI Healthcare Assistant using the Hugging Face Transformers library and Prompt Engineering techniques. The chatbot is designed to provide structured and consistent responses to medical queries while demonstrating best practices in prompt design. The project focuses on improving response quality, reducing hallucinations, and generating well-formatted outputs suitable for different healthcare scenarios.

Note: This project is developed for educational purposes only and is not intended to replace professional medical advice or diagnosis.

2. Model Selection

The chatbot uses TinyLlama/TinyLlama-1.1B-Chat-v1.0, an instruction-tuned language model available on Hugging Face.

TinyLlama was selected because it is lightweight, easy to deploy, and performs efficiently on Google Colab without requiring high-end hardware. Although larger models generally provide stronger reasoning capabilities, TinyLlama offers a good balance between performance, response quality, and computational requirements, making it suitable for academic projects.

The implementation uses the following Hugging Face components:

- AutoTokenizer
- AutoModelForCausalLM
- Transformers Pipeline

These components simplify loading the model, tokenizing user input, and generating responses through a unified interface.

3. Prompt Engineering Techniques Used

To improve the chatbot's response quality, several prompt engineering techniques were applied throughout the project.

Role Prompting

The model is assigned the role of an AI Healthcare Assistant before answering questions. This encourages consistent and professional responses aligned with healthcare applications.

Context Prompting

Each prompt provides sufficient context about the user or patient scenario. Adding context reduces ambiguity and helps the model generate more relevant responses.

Constraints

Prompts include explicit instructions such as avoiding unsupported assumptions, not providing definitive diagnoses, and responding only with available information. These constraints help reduce hallucinations and improve reliability.

Few-Shot Prompting

Three example input-output pairs are provided before the actual query. This teaches the model the desired response format and improves consistency when summarizing clinical notes.

Chain-of-Thought Prompt Design

Instead of asking the model to reveal its reasoning, the prompt instructs it to analyze the information internally before producing a concise clinical assessment and recommendation. This follows current prompt engineering best practices.

Structured Output

The chatbot is instructed to return information in JSON format when required. Structured outputs make it easier for downstream software systems to process the generated information.

4. Challenges Faced

Several challenges were encountered during development:

- Selecting a language model that could run efficiently on Google Colab.
- Preventing the model from generating unsupported medical information.
- Designing prompts that consistently produced structured responses.
- Ensuring that JSON outputs followed the required format without additional text.
- Maintaining response consistency across different prompt styles and healthcare scenarios.

These challenges were addressed through careful prompt design and by using an instruction-tuned language model.

5. Improvements After Prompt Refinement

The chatbot's performance improved significantly after applying prompt engineering techniques.

The original prompts were short and ambiguous, often resulting in inconsistent or unstructured responses. After introducing role prompting, contextual information, constraints, and structured output formatting, the responses became more accurate, organized, and easier to understand.

Few-shot prompting further improved the model's ability to convert clinical notes into structured summaries, while JSON formatting enabled machine-readable outputs for healthcare applications.

Overall, prompt refinement reduced hallucinations, improved consistency, and enhanced the usability of the chatbot.

6. Conclusion

This project successfully demonstrates the development of an AI Healthcare Assistant using Hugging Face Transformers and Prompt Engineering techniques. The chatbot supports medical question answering, structured clinical summaries, JSON-based outputs, and role-specific responses while using a single instruction-tuned language model.

The project highlights the importance of prompt engineering in improving the quality, consistency, and reliability of responses generated by Large Language Models. Future enhancements may include Retrieval-Augmented Generation (RAG), integration with electronic health records, support for multiple language models, and deployment as a web-based healthcare assistant.

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